# A Multiview Difference Feature Perception Network for Change Detection in Bi-Temporal Remote Sensing Images

Lanxue Dang, Shilong Li, Xiao Wang, *Member, IEEE*, and Shenshen Li

***Abstract*—Perceiving difference features from bi-temporal remote sensing images is one of the critical steps in deep learning-based change detection (CD) models. Current methods usually fuse features of bi-temporal images from a single view at a fixed stage. However, they face two key challenges: 1) the bias of a single view leads to inadequate capture of change information and 2) the fusion strategy in the fixed stage ignores the potential correlation between the high-level semantic features and the underlying detailed features. To address these limitations, this article proposes a multiview difference feature perception network. It captures the essential difference information from various perspectives via the multiview perception module (MVPM), which runs through the encoding and decoding stages. The MVPM consists of three stages: interaction, extraction, and fusion. Specifically, the multiview interaction convolution operator is designed to obtain a unified representation of the multiview difference features during the interaction phase. Then, in the extraction stage, the unified representation is continuously decomposed into multiview difference features through the feature decoupling module (FDM). Subsequently, the multiview fusion module (MVFM) is constructed during the fusion stage, and this module can enhance the ability of neural networks to recognize change information by aggregating the multiview difference features of different receptive fields. Furthermore, an independent fusion module is designed at the end of the network to enhance the network's perception of various scale change targets. Extensive experiments on three publicly available CD datasets validate the effectiveness of the proposed method.**

***Index Terms*—Change detection (CD), feature perception, information interaction, multiview, remote sensing.**

## I. Introduction

CHANGE detection (CD) aims to identify differences between bi-temporal remote sensing images [1]. It is a powerful tool for conducting bi-temporal analysis tasks such as monitoring land use changes, urban expansion, disaster assessment, deforestation, and other similar processes. It is highly significant in remote sensing [2].

Traditional CD methods primarily rely on statistical techniques such as change vector analysis [3] and Markov random field [4] models. However, these methods depend on manually constructed feature representations and have limited capacity to model complex high-level change information. Data-driven artificial intelligence's success has shifted CD development from traditional methods to deep learning. However, the limitations in perceiving difference features between bi-temporal remote sensing images by existing CD methods restrict their ability to detect changes. Then, we analyze their limitations from two aspects: fusion methods and stages.

The fusion methods of bi-temporal features are typically classified into three categories. The first type is the "Diff" operation, illustrated by WNet [5], USSFC-Net [6], and AFDE-Net [7] as examples. A single "Diff" operation is better at detecting changes with significant color differences, but it cannot discern specific change targets, like "different objects which have the same spectrum" or "the same objects have different spectrum." The second type is "Concat" operations, as seen in AERNet [1], SwinSUNet [8], and TransUNetCD [9]. For instance, using a single "Concat" operation provides a more comprehensive representation of the bi-temporal features but may lack an understanding of the change information. The third type is the "Add" operation used in GVA-CD [10], employing a single "Add" operation as the standard feature fusion method, which can retain feature information. Nevertheless, it can also introduce a considerable amount of redundant information when capturing the change features in complex scenes. While these methods have produced favorable outcomes, the networks usually employ a single operation to extract change information from bi-temporal features, limiting the identification of change regions.

At the stage of fusion bi-temporal features, deep learning-based CD methods can be divided into early merging and Siamese structures. The early fusion structures typically adopt a single-branch design, where bi-temporal images are fused and fed into the network for CD [Fig. 1(a)]. For example, FC-EF [11] is a classic CD network based on an early fusion structure, which concatenates bi-temporal images for feature extraction at the encoder stage. By contrast, the Siamese

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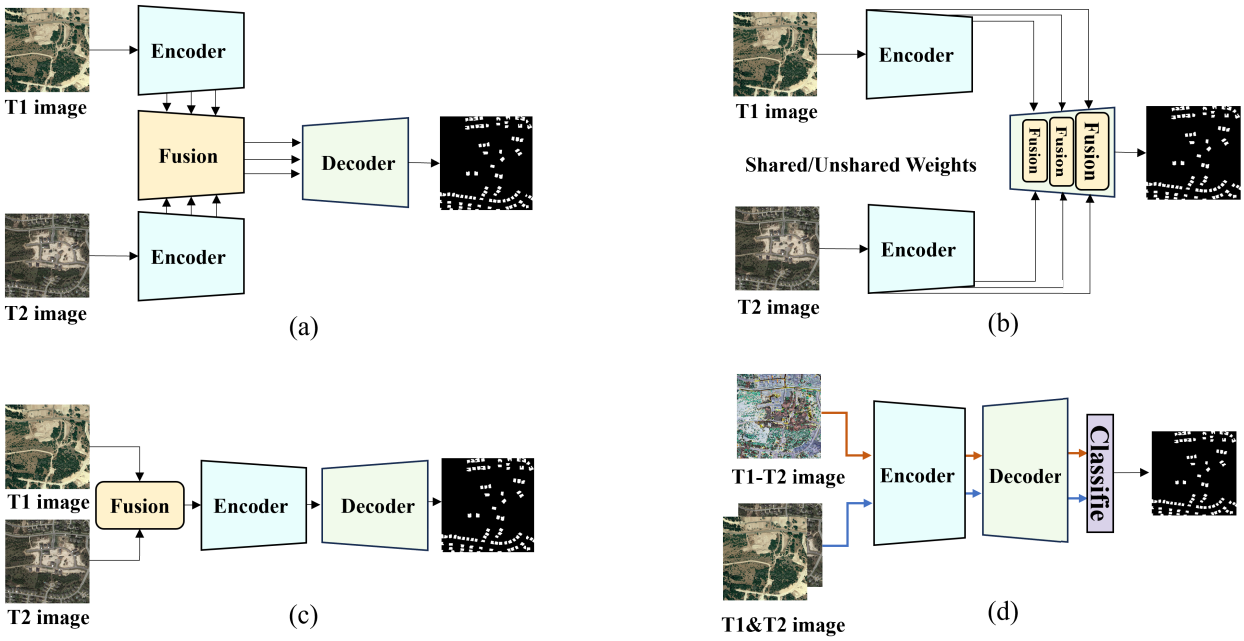


Fig. 1. (a) CD network based on early fusion structure. (b) CD network based on Siamese encoding fusion structure. (c) CD network based on Siamese decoding-fusion structure. (d) Basic architecture of MVU-Net.

structure refers to a dual-encoder–single-decoder structure with shared (or unshared) weights. According to the capture phase of the change information, networks based on the Siamese structure can be categorized into two types. The first type is focused on the acquisition in the encoding phase, which successively fuses the bi-temporal features of different scales and sends them to the decoding part simultaneously [Fig. 1(b)]. For instance, the ChangeFormer [12] employs the shared Transformer as a bi-temporal encoder, fuses the features through the difference module, and utilizes the lightweight multilayer perceptron (MLP) decoder for CD. The second type of Siamese-based network extracts change information during the decoder stage, which incorporates bi-temporal features into the decoder via skip connections, as illustrated in Fig. 1(c). For example, networks such as FC-Siam-conc [11] and FC-Siam-diff [11] progressively incorporate bi-temporal encoding attributes with the forward propagation of the network in the decoding phase through concatenation and subtraction, respectively. Nevertheless, the network's perception of change information is inadequate when change features are fused at a single stage, such as the encoding or decoding stage in the U-shaped network. This is primarily due to the fact that the network's learning of the change information does not occur continuously.

To address the challenges above, this article proposes a multiview difference feature perception U-shaped structure network (MVU-Net), the structure of which is illustrated in Fig. 1(d). MVU-Net employs a novel encoding method for fusing multiple sets of bi-temporal features to generate different features from different viewpoints. Subsequent information complementarity multiview difference features across the whole stage avoids the incomplete perception of complex changing targets by a single-view difference feature at a fixed stage. In particular, MVU-Net accomplishes the interaction, extraction, and fusion of multiview difference features via the multiview perception module (MVPM). The module consists of three phases. First, the MVU-Net proposes a multiview interaction convolution operator based on the traditional convolution operator in the interaction stage. This operator improves the single input to the multiview feature input and obtains a unified representation of the multiview difference feature, as illustrated in Fig. 2. Second, in the extraction stage, the MVU-Net designs a feature decoupling module (FDM) that iteratively decomposes the unified representation into multiview difference features, which are subsequently fed into the next stage. Third, during the fusion stage, the multiview fusion module (MVFM) enhances the network's capacity to perceive change information by aggregating the multiview difference features of different receptive fields. Furthermore, a bottom-up multiscale feature progressive fusion module (MFPFM) was incorporated after the network to enhance the utilization of multiview difference features and facilitate the perception of varying scale change targets. The main contributions of this article are as follows.

1) This article proposes a multiview difference feature perception U-shaped structure network as a solution to the limitation of change recognition ability in complex scenes when using a single view or fixed stage for fusing features of bi-temporal images in CD tasks. MVU-Net utilizes the interaction between multiview difference features for encoding and decoding, enabling it to integrate critical information across multiview features and accurately perceive complex change regions in bi-temporal remote sensing images.
2) We propose an MVPM to capture the critical difference information from different perspectives. The module consists of three stages: interaction, extraction, and

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fusion. We design a multiview interaction convolution operator, an FDM, and an MVFM for different stages. The rich changing information is perceived through unified representation, decoupling, and aggregation of multiview features.

3) We design a bottom-up MFPFM, which fuses multiscale change information in the decoder stage to enhance the network's ability to perceive change targets of varying scales.

The remainder of this article is structured as follows. Section II provides a brief overview of previous related studies. Section III presents comprehensive details of the multiview difference feature perception U-shaped structure network. Section IV shows our experimental setup, and Section V gives detailed experimental results and discussion. Finally, Section VI concludes this article.

## II. Related Work

CD aims to detect differences between images taken at different times. Traditional CD methods rely primarily on extracting spectral information from images to identify changes, such as change vector analysis [3], principal component analysis [13], and multivariate CD [14]. Despite their importance in CD, these techniques often require setting thresholds from experts in the decision-making process, which is time-consuming and requires a high level of expertise, thereby reducing efficiency. As a result, machine learning algorithms such as support vector machines [15], decision trees [16], and random forests [17] have been used in remote sensing image CD due to their adaptability and flexibility. These algorithms can handle complex data patterns, improving detection accuracy. However, they frequently depend on manually crafted features, limiting their ability to capture advanced features and potentially impacting overall performance.

In recent years, deep learning techniques have attracted increasing attention in remote sensing image CD to overcome the limitations associated with traditional methods. The application of deep learning methods enables the automatic extraction of advanced feature representations directly from raw data, thereby reducing the need for human intervention and enhancing the accuracy and efficiency of detection. In the following, we review the related work on deep learning-based CD methods from two perspectives: the fusion methods and stages of bi-temporal features.

### A. Fusion Methods of Different Bi-Temporal Features in the CD Network

As the fundamental step of extracting change information, the fusion method of bi-temporal features is a widely employed approach among scholars, with the most commonly utilized techniques being "Diff," "Concat," and "Add."

The "Diff" operation extracts pixel-level changes through point-by-point differences. For example, the AFDE-Net [7] employs a point-by-point difference at the outset of the ensemble spatial–channel attention fusion module, thereby learning more about the spatial and spectral information of the changing region. The USSFC-Net [6] uses the absolute value of the point-by-point difference results to fuse the bi-temporal encoding features to obtain a richer feature map of the change object. The WNet [5], which also uses point-by-point difference absolute value operation, is fused at the head of the difference enhancement module to highlight the highly divergent changes in the bi-temporal features diagram. However, such operations are more adept at detecting significant color contrasts and have exacting standards for the imaging quality of bi-temporal images. Consequently, they are not particularly responsive to the change targets of "different objects which have the same spectrum" or "the same objects have different spectrum."

The "Concat" operation integrates features by channel-wise concatenation, preserving bi-temporal information. For example, AERNet [1] gradually concatenates the layered bi-temporal encoding features and the forward propagation features through a skip connection to help the network sense and refine the edge of the changing region. In addition, TransUNetCD [9] concatenates the CNN encoding features of the double steam and divides the concatenate features into sequential patches to capture the global context information in the Transformer-based encoder to enhance the depth feature representation. Similarly, SwinSUNet [8] concatenates the sequential patches of bi-temporal to restore the spatial resolution of the change features gradually. However, although the "concat" operation can completely model the feature of each temporal, it lacks the understanding of the change information.

The "Add" operation fuses features via point-wise addition, enhancing spatial consistency. For example, GVA-CD [10] adds a point-by-point at the head of the geometric variation adaptive modules to build a contextual point set, which is used to calculate the difference measurement between bi-temporal features. However, the "Add" operation, as a standard feature fusion method, can better retain feature information. However, it will lead to a large amount of redundant information when capturing the changing features of complex scenes.

Therefore, this article proposes a multiview difference feature perception U-shaped structure network to maintain the advantages of different fusion methods. The network can integrate the critical difference features from multiviews and perceive the complex change areas in remote sensing bi-temporal images.

### B. CD Network Based on Early Fusion Structure

The early fusion structure usually uses the form of a single branch to design the CD network. This kind of network first fuses the bi-temporal image and then uses a network similar to semantic segmentation for CD. FC-EF [11] directly concatenates bi-temporal images for feature extraction in a manner that disregards the integrity of the independent information in the original image. Furthermore, using a single-branch approach fails to leverage the temporal relationship of bi-temporal features, which is detrimental to the CD process. A detailed account of the process of minimizing the loss function in the single-branch CD network is illustrated in the following equation:

$$\min_{\theta} L\left(F_{\mathrm{de}_\theta}\left(F_{\mathrm{en}_\theta}\left(\mathrm{Concat}\left(T_1, T_2\right)\right)\right), Y\right) \tag{1}$$

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where $F_{\mathrm{en}_\theta}$ and $F_{\mathrm{de}_\theta}$ denote the encoder process and decoder process, respectively; Concat denotes the concatenate operation; $L$ denotes the loss function; $T_1$ and $T_2$ denote the bi-temporal features, respectively; $Y$ denotes the real labels of the changing region; and $\theta$ denotes the learnable parameters of the model. For the bi-temporal images, the concatenate is performed first. Then, it goes through the feature encoding and decoding stages sequentially, and finally, the loss learning is performed at the real labels.

### C. CD Network Based on Siamese Structure

Siamese CD networks adopt a dual-branch structure, utilizing encoders with shared or nonshared weights for feature extraction. Based on the stage of change information capture, they are categorized into Siamese encoding-fusion networks and Siamese decoding-fusion networks.

The Siamese network fuses bi-temporal features during the encoding phase, with innovations focusing on fusion strategies, attention mechanisms, and Transformer integration. First, ChangeFormer [12] uses a shared Transformer as the bi-temporal encoder and a lightweight MLP decoder, with feature fusion via a difference module. Similarly, MAFGNet [18] extracts coarse change features through a difference module and leverages a graph convolution decoder to guide self-attentional learning. To utilize multilevel encoded features, MFCEN [19] proposes a band-wise fusion strategy and temporal attention module to enhance change region focus, while BTAF [2] adopts a hybrid attention module for progressive multilevel feature filtering. For temporal information utilization, Eatder [20] mines time clues via a full-range fusion module, and FDINet [21] captures spatiotemporal correlations using a wavelet-based interaction module. In Transformer-based designs, Siamixformer [22] employs Transformers for feature fusion, and DiFormer [23] introduces a token exchange-based difference module to highlight bi-temporal discrepancies.

The Siamese decoding-fusion structure is characterized by dual encoders and a single decoder; these networks transmit bi-temporal features to decoders via skip connections, with research focusing on skip connection optimization, decoder module enhancement, and other combinations. First, for skip connection improvements, FC-Siam-Conc and FC-Siam-Diff [11] use concatenation or difference for fusion during decoding. SNUNet-CD [24] employs dense skip connections to preserve high-resolution features and reduce edge uncertainty or small-target omission. SNUNet3 + [25] integrates spatial-semantic information via full-scale skip connections. Second, in the network improvement of the more effective decoding module, AERNet [1] introduces global context aggregation and edge refinement modules. HANet [26] refines spatiotemporal features to capture row and column dependencies to boost robustness. Finally, some scholars have combined the structural design of the skip connection and the decoding module to enhance the extraction of change information in a CD network. To address the issue of loss of spatial information resulting from downsampling, which leads to incomplete buildings and irregular output boundaries, AFDE-Net [7] adds spatial–channel attention in skip connections and deep supervision in decoders to mitigate spatial information loss. Conversely, CSAGC [27] uses contextual spatial awareness and spatial integration to enhance context comprehension.

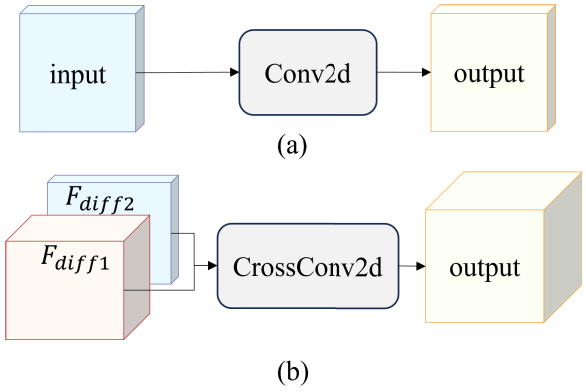


Fig. 2. Visualization of (a) traditional convolution operator and (b) multiview interaction convolution operator.

Nevertheless, although the Siamese network based on two branches retains the target features of the image before and after the bi-temporal, it does not fully use the change information. This results in the edge detection of the change region not being optimal, leading to an increase in false positives and missed detections. Similarly, we present a summary of the process of minimizing the loss function of the structure, as shown in the following equation:

$$\min_{\theta} L\left(F_{\mathrm{de}_\theta}\left(Fu\left(F_{\mathrm{en0}_\theta}(T_1), F_{\mathrm{en1}_\theta}(T_2)\right)\right), Y\right) \tag{2}$$

where $F_{\mathrm{en0}_\theta}$ and $F_{\mathrm{en1}_\theta}$ denote the encoding part of the network, the weights of which can be shared or learned separately. Fu is a fusion operation for bi-temporal encoded features; $L$ denotes the loss function; $T_1$ and $T_2$ denote the bi-temporal features, respectively; $Y$ denotes the true labels of the changing regions; and $\theta$ denotes the learnable parameters of the model. For the bi-temporal images, independent feature encoding is performed first; then, feature fusion is performed; and finally, the fused features are fed into the decoding network for loss learning at the true labels.

## III. Method

This section begins with an overview of the general structure of the MVU-Net, followed by a detailed examination of the MVPM and the MFPFM. Finally, the loss function employed for training the network is presented.

### A. Overall Architecture

The architecture of MVU-Net is depicted in Fig. 3. It adopts an overall single-branch U-shaped design, consisting of a single-branch encoder, a single-branch decoder, and a multiscale feature classifier. MVU-Net uses its encoder to extract hierarchical features through the progressive reduction of spatial resolution for multiview difference features. This is followed by upsampling in the decoding part, which reduces the deep semantic features to shallow texture features. Finally,

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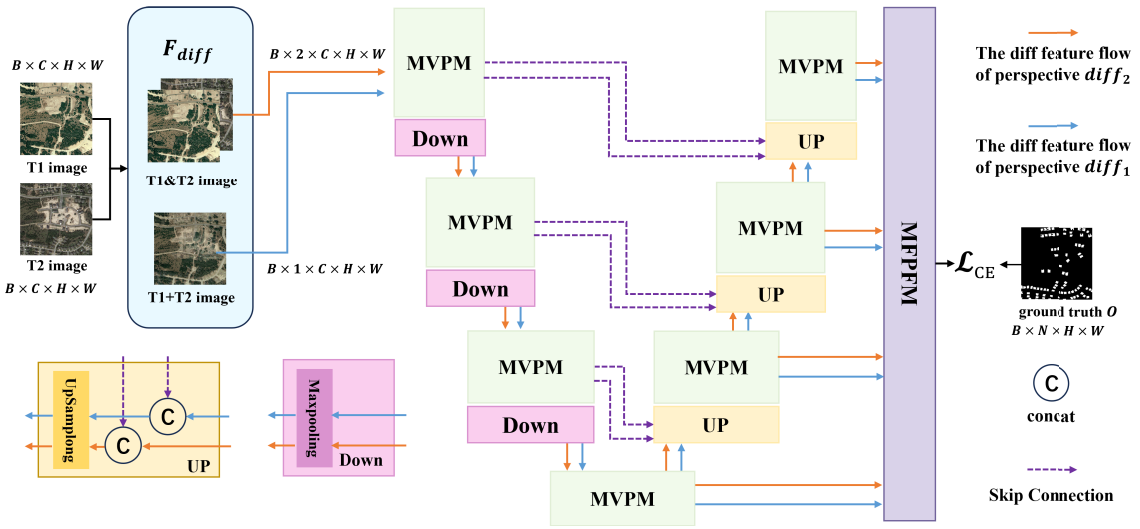


Fig. 3. Overall architecture of MVU-Net. $F_{\text{diff}}$ represents the generation process of multiview difference features, MVPM is our proposed MVPM, and MFPFM represents the multiscale feature progressive fusion module for intensive prediction. The UP module represents the upsampling implementation details of the different features, and the Down module represents the downsampling details of the different features. In addition, the orange line segment flow and the blue line segment flow, respectively, represent the flow direction of different features in the network, while the purple dotted line represents the feature flow direction of the skip connection part.

the features are transmitted to the classifier for dense prediction. Furthermore, skip connections integrate encoder features into the decoder, thereby compensating for the texture detail information of earlier features in the deeper features. The orange line segment flow and the blue line segment flow in Fig. 3 illustrate the flow of different features in the network, respectively, while the purple dashed line indicates the flow of features in the skip connection part.

The MVU-Net receives bi-temporal remote sensing images, $T_1$ and $T_2$, which are represented in the form of matrices of dimensions $\mathbb{R}^{B\times C\times H\times W}$; $B$ denotes the number of batches of network inputs; $C$ denotes the feature channel dimension; and $H$ and $W$ denote the length and width of the feature map, respectively. The MVU-Net then outputs a change prediction map $O \in \mathbb{R}^{B\times N\times H\times W}$, where $N$ denotes the number of categories of predicted outcomes. The bi-temporal remote sensing images utilized for CD are initially employed to generate the multiview difference features $\text{diff}_1$ and $\text{diff}_2$ by concat as well as add, whose feature map sizes are $\text{diff}_1 \in \mathbb{R}^{B\times F_1\times C\times H\times W}$ and $\text{diff}_2 \in \mathbb{R}^{B\times F_2\times C\times H\times W}$, where $F_1$ and $F_2$ denote the different dimensions. The rationale for selecting these two different feature generation methods is as follows. The Concat operation, a common method for feature fusion, concatenates the bi-temporal features along the channel dimension $C$, thus ensuring the preservation of complete image information. In contrast, the Add operation belongs to the parameter-free processing method, which effectively highlights the change region and provides change information.

Subsequently, the multiview difference features generated earlier are jointly fed into the encoder. The encoder consists of four MVPM blocks, followed by three downsampling modules that progressively reduce the spatial resolution of features. The MVU-Net employs a symmetrical structure design, with the decoding component comprising three MVPM blocks and three up modules, which restore the spatial resolution of the features gradually. The network ultimately inputs the multilevel decoding features into MFPFM, where they are processed to yield pixel-level detection results in the form of an image of dimensions $O \in \mathbb{R}^{B\times N\times H\times W}$. For the skip connections, a simple concatenation operation is used, thereby eliminating the need to introduce additional parameters.

Compared to previous works, MVU-Net enhances the utilization efficiency of change information between different temporal features. The network employs temporal phase change features in both the encoder and decoder phases and effectively overcomes the loss of detail information that may result from traditional fixed-position feature fusion by adopting a multiview different feature interaction strategy. Specifically, MVU-Net constructs different features at the input stage using multiple methods, thus enabling the network to capture critical feature information from different perspectives more comprehensively and to learn to identify critical regions of change. Furthermore, we train the network by minimizing the loss function, as shown in the following equation:

$$\min_{\theta} L\left(F_{\text{de}_\theta}\left(F_{\text{en}_\theta}\left(F_{\text{diff}}^{\text{concat}}, F_{\text{diff}}^{\text{add}}\right)\right), Y\right) \tag{3}$$

where $F_{\text{diff}}^{\text{concat}}$ and $F_{\text{diff}}^{\text{add}}$ denote the difference features from different perspectives, which are generated by the bi-temporal features using concat and add, respectively; $L$ denotes the loss function; $Y$ denotes the true labeling of the change region; and $\theta$ denotes the learnable parameters of the model.

### B. Multiview Perception Module

In the case of remote sensing images of the exact location and different periods, image differences often play a better guiding role for network segmentation. Consequently, many researchers have proactively investigated the fusion strategy of bi-temporal features for image differences. However, the

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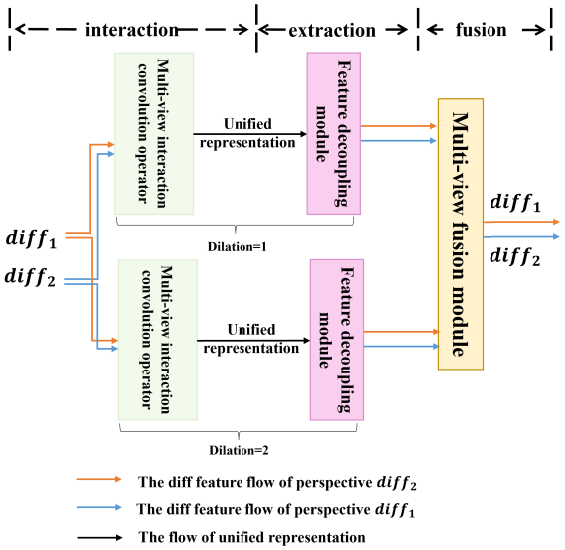


Fig. 4. Detailed structure of the MVPM, which is composed of interaction, extraction, and fusion stages. Among them, interaction and extraction are parallel structural designs, and the difference between parallel branches is the hole factor of convolution.

**Algorithm 1** Implementation Steps of MVPM

**Input:** $\text{Diff}_1, \text{Diff}_2$ (Dual flow of perspective feature)
**Output:** $\text{Diff}_1^{\text{out}}, \text{Diff}_2^{\text{out}}$
**//step1:** Obtain unified representation of different receptive fields through MVICony.
$\text{Unified}_{\text{dil=1}} = \text{MVIConv}_{\text{dil=1}}(\text{Diff}_1, \text{Diff}_2)$
$\text{Unified}_{\text{dil=2}} = \text{MVIConv}_{\text{dil=2}}(\text{Diff}_1, \text{Diff}_2)$
**//step2:** Decoupled unified representation through FDM.
$\text{Diff}_1^{\text{dil=1}}, \text{Diff}_2^{\text{dil=1}} = \text{MFD}(\text{Unified}_{\text{dil=1}})$
$\text{Diff}_1^{\text{dil=2}}, \text{Diff}_2^{\text{dil=2}} = \text{MFD}(\text{Unified}_{\text{dil=2}})$
**//step3:** Fusion features of different receptive fields through MVFM.
$\text{Diff}_1^{\text{out}} = \text{MVFM}(\text{Diff}_1^{\text{dil=1}}, \text{Diff}_1^{\text{dil=2}})$
$\text{Diff}_2^{\text{out}} = \text{MVFM}(\text{Diff}_2^{\text{dil=1}}, \text{Diff}_1^{\text{dil=2}})$

MVPM eschews the fusion of bi-temporal features and instead opts for the interaction of features from different perspectives to discern the change in information, where different perspectives refer to the use of bi-temporal features to merge different strategies according to the different information of the images. The MVPM consists of interaction, extraction, and fusion stages. Its structure is shown in Fig. 4.

According to the design phase of MVPM, the multiview interaction convolution operator, FDM, and MVFM are introduced. We summarize the MVPM workflow in Algorithm 1. As the core phase of MVPM, the interaction phase learns shared change information from multiview change features and generates a unified representation of these features. By contrast, the extraction stage decouples the unified representation of commonalities into multiperspective features, providing continuous updating and learning for the subsequent development of the network. The fusion stage involves the interaction of different void factors, the fusion of extracted features, and the perception of temporal and spatial relationships between changes in dual-temporal remote sensing images and features from the feature space of different receptive fields.

*1) Multiview Interaction Convolution Operator:* The initial stage of the MVPM workflow corresponds to the interaction phase. To improve the ability of MVU-Net to characterize the change information of bi-temporal features, MVPM designs a multiview interaction convolution operator (MVIConv), which utilizes the change features from different views to learn interactively. Fig. 5 shows the primary process of this operator. In the following, we elaborate on it from two perspectives: its theoretical basis and workflow.

The MVIConv relies on the following theoretical framework. First, the parameter reuse strategy inherits the successful experience of Siamese networks in CD, aligns the dual-phase feature distribution by forcing the encoder to share weights, and suppresses transient interferences such as lighting and season. Second, the intrinsic permutation invariance property of the operator derives from satisfying its mathematical equation

$$\text{MVIConv}(D_1, D_2; \theta_z) = \text{MVIConv}(D_2, D_1; \theta_z). \quad (4)$$

This allows the network to maintain detection stability without relying on temporal information. The multiview interaction mechanism implicitly constructs feature correlation channels through shared convolutional kernels, and its mathematical modeling can be expressed as follows:

$$\text{MVIConv}(D_1, D_2; \theta_z) = \text{Conv}(R(D_1) \| R(D_2); \theta_z) \quad (5)$$

where $D_1$ and $D_2$ represent the changing features of different perspectives and $\theta_z$ represents the multiplexed convolution weights. $\|$ is a cascade operation along the feature dimension. $R$ represents the dimension transformation. Finally, Fig. 6 shows the visualization analysis of the characteristic heat map before and after the MVIConv operator. Comparisons show that, relative to single-perspective heatmaps (second and third columns), the unified representation learned by MVIConv (fourth column) highlights change regions more clearly and accurately and is more consistent with the ground truth in the last column. This intuitively suggests that MVIConv can effectively integrate multiview features, suppress irrelevant information, improve the representational ability of key change areas, and thus improve the accuracy of CD.

For the inputs of multiview difference features $\text{diff}_1^i \in \mathbb{R}^{B\times F_1\times C_1^i\times H_i\times W_i}$ and $\text{diff}_2^i \in \mathbb{R}^{B\times F_2\times C_2^i\times H_i\times W_i}$, where $i$ denotes different levels of MVPM ($i = 1, 2, 3, 4$ for encoding stage and $i = 5, 6, 7$ for decoding stage), MVIConv first aligns different features using the repeat operation to generate feature dimension representations for $\text{diff}_1^i \in \mathbb{R}^{B\times F_1\times F_2\times C_1^i\times H_i\times W_i}$ and $\text{diff}_2^i \in \mathbb{R}^{B\times F_1\times F_2\times C_2^i\times H_i\times W_i}$, respectively. Then, the multiview difference features are fused in the channel dimension by a concatenation operation to generate the fusion feature $\mathbb{R}^{B\times F_1\times F_2\times (C_1^i+C_2^i)\times H_i\times W_i}$. Consequently, features as mentioned earlier are subjected to a reshaping process, whereby the different dimension is integrated with the batch dimension to generate cross-feature $\mathbb{R}^{(B\times F_1\times F_2)\times (C_1^i+C_2^i)\times H_i\times W_i}$. The cross-feature is then sequentially fed into the normal Conv2d operator, which is permutation invariant with respect to the crossover features. This is because the Conv2d operator

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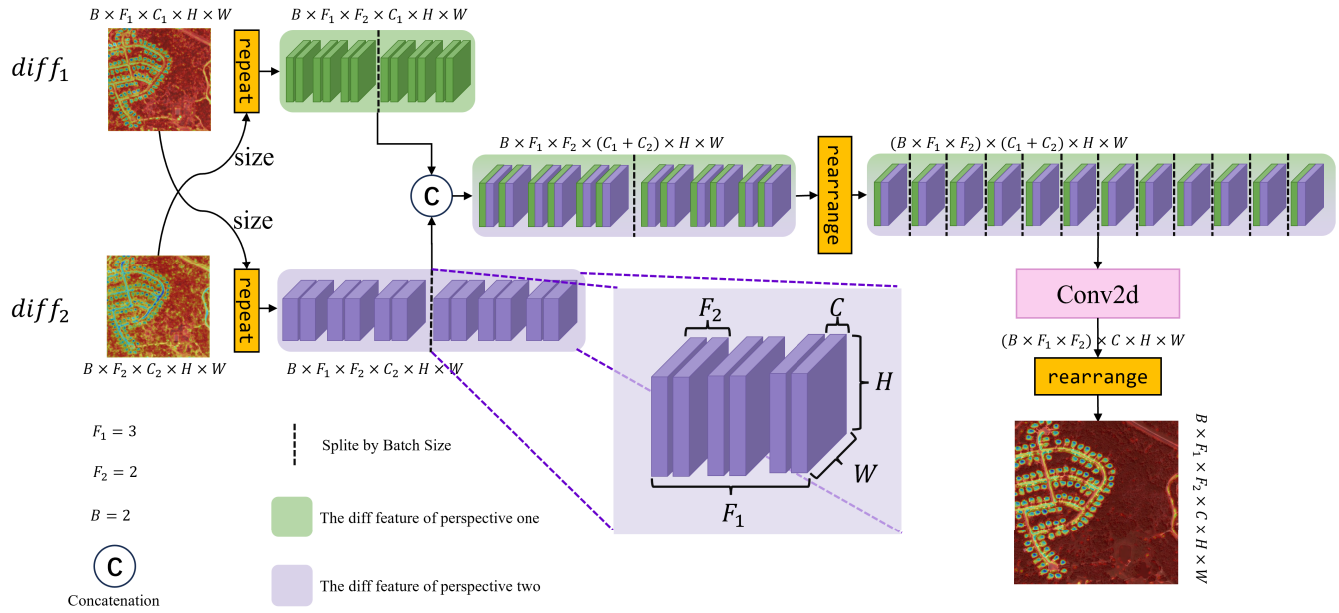


Fig. 5. Structure of multiview interaction convolution operator. The visualization uses the examples of $F1 = 3$, $F2 = 2$, and $B = 2$. The specific implementation of the operator is related to the different features of the selected area. Detailed structure of the different features after alignment is shown.

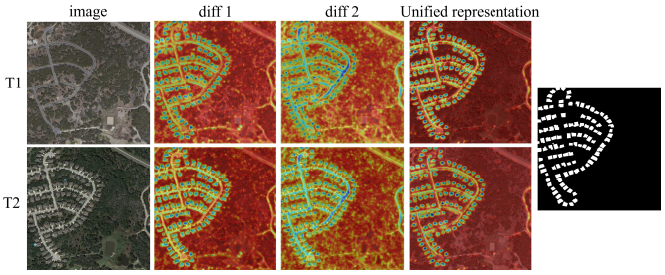


Fig. 6. Visualization effect of heatmap based on MVIConv features.

weights are shared. The operator then shapes the output features into the unified representation $\mathbb{R}^{B\times F_1\times F_2\times C^i\times H_i\times W_i}$.

*2) Feature Decoupling Module:* In the extraction phase of the MVPM, an FDM was designed to decompose the unified representation into multiview difference features. This design enables the continuous updating and learning of the multiview difference features within the network, and the structure of this FDM is illustrated in Fig. 7. The blue and orange solid lines represent the flow of features from different views, respectively. The black flow line represents the feature flow of the unified representation.

FDM averages the unified representation in the different dimensions, enabling the unified representation to interact with the features through their averaged representations. The regular operations in the image feature target 4-D features. However, the homogenized MVIConv has five dimensions $\mathbb{R}^{B\times F_2\times C^i\times H_i\times W_i}$. Therefore, FDM shapes the different dimension $F_2$ into the $B$ dimension to generate $\mathbb{R}^{(B\times F_2)\times C^i\times H_i\times W_i}$. The features are then passed through successive BN-ReLU-Conv2d-BN-ReLU operations. This shaping enables the network to learn more general feature representations with shared weight parameters. Then, the dimension is restored to generate the representation of the different features to generate $\text{diff}_2^i \in \mathbb{R}^{B\times F_2\times C^i\times H_i\times W_i}$. The feature $\text{diff}_1^i$ is generated by a continuous Conv2d-BN-ReLU combination. In contrast to the generation of the feature $\text{diff}_2^i$, the coupling and decoupling operations of the feature $\text{diff}_1^i$ will act on the feature different dimension $F_1$ and the channel dimension $C^i$.

*3) Multiview Fusion Module:* The MVFM is designed for the final stage. Within this module, it first fuses the multiview difference features from the parallel branches of the previous stage. Additionally, it perceives the spatiotemporal relationships between the change information of bi-temporal remote sensing images and their corresponding features, leveraging feature spaces with different receptive fields. The specific structure of this module is shown in Fig. 8, where the blue and orange arrows indicate the flow of different views and features, respectively.

The MVFM inputs, $\text{diff}_1^i \in \mathbb{R}^{B\times F_1\times C_1^i\times H_i\times W_i}$ and $\text{diff}_2^i \in \mathbb{R}^{B\times F_2\times C_2^i\times H_i\times W_i}$, are initially processed through two parallel branches of interaction and extraction phase, which differ in the number of dilated ratios of the corresponding convolution operators. Each branch outputs the difference features of the multiviews, which are subsequently superimposed on the different features of the corresponding viewpoints in a sum. This operation is illustrated in the following equation:

$$\begin{aligned}\left(f_1^i, f_2^i\right) &= \text{branches}_{d=1}\left(\text{diff}_1^i, \text{diff}_2^i\right)\\ \left(g_1^i, g_2^i\right) &= \text{branches}_{d=2}\left(\text{diff}_1^i, \text{diff}_2^i\right)\\ \text{Cross } F_1^i &= f_1^i + g_1^i\\ \text{Cross } F_2^i &= f_2^i + g_2^i. \end{aligned} \tag{6}$$

where $d$ denotes a different dilated ratio. Subsequent convolutional combinations are employed to extract a more extensive receptive field, thereby enhancing the capacity of the features to discern local information. Specifically, the above features extracted by different branches are initially fused using a

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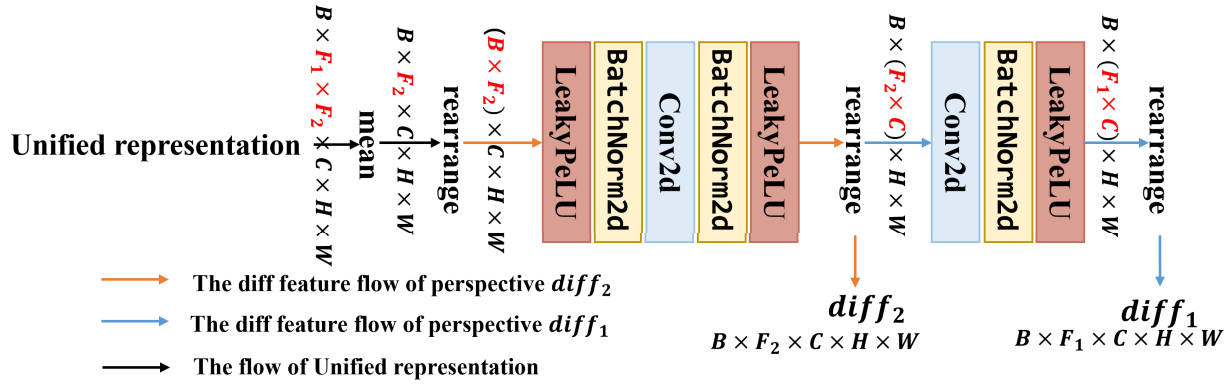


Fig. 7. Structure of the FDM. The feature flow direction of different views is represented by blue and orange solid lines, and the black flow line represents the feature flow of unified representation.

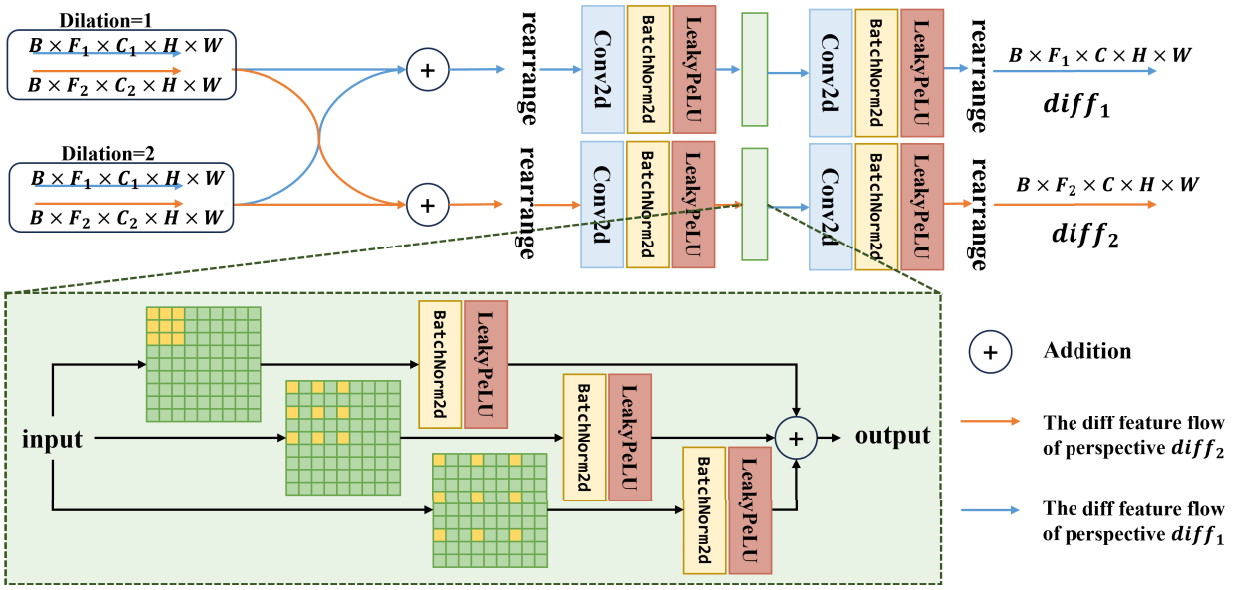


Fig. 8. Structure of the multiview interaction module. The blue and orange arrows indicate the flow direction of different visual angles of different features.

$1 \times 1$ convolutional combination. This is followed by applying three $3 \times 3$ convolutional combinations in parallel, which extract different scales of the receptive field. A multiscale feature extraction process captures contextual information at multiple scales. Finally, the parallel processed branches are fused by summation, and the final module output is obtained using a $1 \times 1$ convolutional combination. The convolutional combination employed in the operations, as mentioned earlier, comprises the Conv2d-BN-ReLU. Each operation preserves the batch dimension and maintains consistency with the feature dimensions involved in coupling and decoupling.

The MVFM employs an interaction of disparate different features for encoding and decoding. This enables the network to synthesize the critical information of various angular features and to learn the key areas of change in the remote sensing bi-temporal images.

### C. Multiscale Feature Progressive Fusion Module

For the U-shaped network, with the deepening of the number of network layers, the channel of the network gradually becomes larger, and the semantic expression ability is also improved. However, this also leads to a reduction in image resolution, resulting in the features obtained after multilayer network convolution ignoring target details. In contrast, shallow networks have fewer feature channels and excel at expressing detailed information, but the semantics of their extracted features are less robust. Solely using the last layer of the feature map for CD can easily lead to small change targets being ignored. Therefore, in this article, a multiscale feature gradual fusion module is designed, which gradually fuses the multilayer decoded features before detection. The design motivation of this module stems from two considerations. First, low-level detail features can provide accurate spatial positional support for high-level semantic features, effectively avoiding positional deviation caused by the lack of detail information in high-level semantic features. Second, advanced semantic features can understand and parse low-level detail features, filtering out misjudgments caused by noise or similar textures in low-level features (e.g., effectively distinguishing between artifacts caused by shadow changes and real ground changes).

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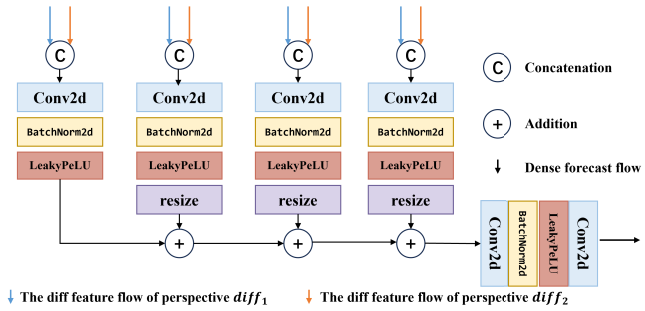


Fig. 9. Structure of MFPFM.

This model promotes the interaction between low-level detail features and high-level semantic features at different levels, retaining the fine localization ability of low-level features and integrating the semantic discrimination ability of high-level features, ultimately achieving high-precision CD and localization in detection tasks.

The structure of the MFPFM is shown in Fig. 9. The module consists of four parallel branches, each starting from Conv2d, which are used for preliminary feature extraction of the input features. Next, the batch normalization layer BatchNorm2d normalizes the convolutional output to ensure the stability of the data distribution, speed up model training, and improve generalization. Subsequently, the LeakyReLU activation function introduces nonlinear transformations to break linear relationships, allowing the model to learn more complex feature representations while mitigating the gradient vanishing problem. Each branch includes a resize operation that adjusts feature map dimensions, ensuring features from different branches are spatially compatible for fusion. The fused features are further refined and optimized by a combination of Conv2d, BatchNorm2d, and LeakyReLU activation functions. The final Conv2d performs dense feature prediction and outputs detection results that have undergone multiscale progressive fusion.

### D. Loss Function

We use the binary cross-entropy (BCE) loss function [16] to calculate the loss during network training. Its definition is as follows:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{M+N}\left(\sum_{i=1 \text{ if } Y=1}^{M} \log\left(\bar{Y}\right) + \sum_{i=1 \text{ if } Y=1}^{M} \log\left(1-\bar{Y}\right)\right) \tag{7}$$

where $M$ represents the number of positive samples, $N$ represents the number of negative samples, $Y$ represents the real label value, and $\bar{Y}$ represents the predicted value.

## IV. Experimental Setup

### A. Datasets

This section describes the LEVIR-CD [28], EGY-BCD [7], and GVLM-CD [29] datasets.

LEVIR-CD is a large-scale building CD dataset obtained from Google Earth. It consists of 20 different regions in many cities in USA, covering the changing characteristics of various buildings, including high-rise apartments, villas, small garages, large warehouses, and more. It contains 637 pairs of bi-temporal high-resolution remote sensing image blocks, with each block sized 1024 × 1024 pixels and a spatial resolution of 0.5 m/pixel.

EGY-BCD is a dataset designed for building CD from high-resolution satellite images, which have a spatial resolution of 0.25 m/pixel. The dataset covers four cities and coastal regions in Egypt, with data collection from 2015 to 2022 via Google Earth. The collection consists of 6091 pairs of small images, each 256 × 256, segmented in a 6:2:2 ratio, with 3655, 1218, and 1218 pairs of images allocated for train, validation, and test, respectively.

GVLM-CD is the first large-scale, open-source very-high-resolution landslide mapping dataset. It includes 17 bi-temporal ultrahigh resolution image pairs with a spatial resolution of 0.59 m obtained through the Google Earth service. Each subdataset contains bi-temporal images and corresponding ground truth maps. Due to the different sizes, shapes, occurrence times, spatial distributions, phenological states, and land cover types of landslide masses in different geographical locations, the image exhibits considerable spectral heterogeneity and intensity differences. This article describes how 7596 sample pairs, each 256 × 256 pixels, are derived from the original images using a nonoverlapping clipping procedure. These samples are divided in a 6:2:2 ratio, resulting in 4558, 1519, and 1519 image pairs for training, validation, and testing, respectively.

Table I shows the details of the three datasets mentioned above. By selecting datasets with different spatial resolutions, data volumes, and sample distributions, we can more comprehensively and objectively verify the robustness and generalization performance of the MVU-Net.

TABLE I
Three Dataset Configurations Used

| Datasets | Spatial Resolution | Size/Image | Number of Images in Datasets | | |
|---|---|---|---|---|---|
| | | | Train | Validation | Test |
| LEVIR-CD | 0.5m/pixel | 1024×1024 | 445 | 64 | 128 |
| EGY-BCD | 0.25m/pixel | 256×256 | 3655 | 1218 | 1218 |
| GVLM-CD | 0.59m/pixel | 256×256 | 4558 | 1519 | 1519 |

### B. Experimental Details

We implement all experiments using Open-CD [30] (a transformation detection toolkit based on PyTorch and Open MMLab). During training, the batch size is set to 8, and the AdamW optimizer uses an initial learning rate of 0.001. We apply random flipping and rotation during training to improve generalization. A comparative experiment was performed on a V100-32GB GPU. The model was trained for 40 000 iterations and reached convergence.

### C. Evaluation Metrics

To assess the performance of the proposed network and compare it with other methods, we choose intersection over union (IoU) and $F1$ as the primary evaluation indicators.

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Meanwhile, we also provide the results of precision and recall that are used to compute $F1$. These metrics are defined as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{8}$$

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

$$\text{Precision} = \frac{\text{TP}}{\text{FP} + \text{TP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

where TP represents the number of true positives, TN represents the number of true negatives, FN represents the number of false negatives, and FP represents the number of false positives.

## V. Result

### A. State-of-the-Art Methods

We used 11 state-of-the-art methods to compare our model. The principles of these methods are briefly described as follows.

1) FC-EF [11] is a single-branch early fusion network that fuses changing features before network learning through "Concat" operations.
2) FC-Siam-conc [11] is a Siamese network in which bi-temporal features are fused using "Concat" operations.
3) FC-Siam-diff [11] is a Siamese network, where two-phase features are fused using "Diff" operations.
4) BiT-CD [31] is a Transformer-based method. First, a Transformer encoder is used to represent the bi-temporal image as a context-rich token. Then, a Transformer decoder is utilized to map the token back to the pixel space in order to refine the original features.
5) ChangeFormer [12] is a Siamese network based on Transformer. It utilizes a layered convolutional encoder to extract multiscale features, followed by a multilayer perceptual decoder to fuse the differences between multiscale features and predict CD labels.
6) Changer-EX [30] proposed a general CD framework. The interactive fusion module is used to alleviate the dislocation problem of dual time images.
7) SNUNet-CD [24] employs the Siamese dense connection structure to improve the method of acquiring feature location information.
8) USSFC-Net [6] is a Siamese-structured network that effectively fuses spatial–spectral features and obtains richer functionalities. Additionally, a multiscale decoupled convolution is designed, which can flexibly capture the multiscale features of changing objects through cyclic multiscale convolutions.
9) HANet [26] introduces a progressive foreground equalization sampling strategy, which helps the model accurately learn foreground image features by initially focusing on them and then gradually incorporating background images. In addition, the HAN module is designed for feature refinement. The module comprises a parallel convolution structure and a channel attention module, where the attention is prioritized to obtain better semantic information.
10) EATDer [20] consists of a Siamese encoder and an edge-aware decoder, which are used to capture basic knowledge and generate change prediction maps from remote sensing images.
11) FEMCD [32] proposes a frequency-enhanced Mamba to address the limitations in capturing features of minor and texture changes due to the lack of frequency information.
12) RSMamba [33] is used for dense prediction tasks in large VHR remote sensing images; it is specifically designed to capture the global context of remote sensing images with linear complexity, facilitating the effective processing of large VHR images.
13) LCCDMamba [34] uses SiamVMamba as a backbone to extract bi-temporal features.
14) CDMamba [35] effectively combines global and local features for handling binary CD tasks.

We implemented these methods using their publicly available code. The hyperparameters of these methods were configured in accordance with their respective original literature.

### B. Quantitative Evaluation

This section presents a quantitative comparison between the state-of-the-art methods and our proposed approach, aiming to substantiate the efficacy of our method. Table II illustrates the quantitative comparison of our method and other methods on the LEVIR-CD, EGY-BCD, and GVLM-CD datasets. The performance of MVU-Net in $F1$ and IoU is always better than that of the competing method. Specifically, it reaches 92.05% and 85.26% in LEVIR-CD, outperforming the best model by 0.85% and 1.82%. In the EGY-BCD dataset, it reaches 81.19% and 68.34%, surpassing the best scores of 1.25% and 1.75%. Finally, in the GVLM-CD dataset, it reaches 89.07% and 80.29%, surpassing the best scores of 0.8% and 1.28%. These results show that MVU-Net has significant performance advantages in the remote sensing CD task, especially in the $F1$ score and IoU key indicators. These indicators are important for remote sensing image CD because they provide a comprehensive evaluation of model performance and balance Precision and Recall. Furthermore, our proposed method yields stable precision and recall rates, and MVU-Net is more effective at balancing missed and false detections.

Overall, these positive results can be attributed to the following three key factors. First, by extracting multiview difference features for CD, MVU-Net can capture more efficient feature representations based on the fundamental characteristics of remote sensing images. Second, MVPM facilitates the utilization of change region information from a bi-temporal feature map and improves the robustness via the interaction of multiview change information. Third, MFPFM is represented by deep to shallow features integrated with multiscale and different perspectives to achieve more accurate detection results.

### C. Qualitative Evaluation

To further investigate the effectiveness of our MVU-Net in detecting complex change targets, three remote sensing image

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TABLE II
COMPARISON RESULTS ON THE LEVIR-CD, EGY-BCD, AND GVLM-CD DATASETS

| Model | LEVIR-CD | | | | EGY-BCD | | | | GVLM-CD | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F1(%) | IoU(%) | Pre.(%) | Rec.(%) | F1(%) | IoU(%) | Pre.(%) | Rec.(%) | F1(%) | IoU(%) | Pre.(%) | Rec.(%) |
| FC-EF [11] | 81.71 | 69.07 | 81.42 | 81.99 | 67.90 | 51.40 | 70.06 | 65.87 | 84.07 | 72.51 | 83.04 | 85.12 |
| FC-Siam-conc [11] | 85.65 | 74.50 | 87.47 | 83.91 | 69.09 | 52.78 | 61.73 | 78.46 | 84.03 | 72.46 | 83.04 | 85.05 |
| FC-Siam-conc [11] | 85.38 | 74.50 | 88.97 | 82.08 | 68.54 | 52.14 | 65.15 | 72.30 | 83.79 | 72.10 | 85.20 | 82.42 |
| BiT-CD [31] | 89.31 | 80.68 | 89.24 | 89.37 | 70.05 | 53.91 | 79.43 | 62.65 | 87.56 | 77.87 | **89.48** | 85.71 |
| ChangeFormer [12] | 90.40 | 82.48 | 92.05 | 88.80 | 77.24 | 62.92 | 77.32 | 77.16 | 87.49 | 77.76 | 88.20 | 86.79 |
| Changer-EX [30] | 90.97* | 83.44* | 92.05* | 89.92* | 79.94 | 66.59 | 83.70 | 76.51 | 87.72 | 78.13 | 89.05 | 86.43 |
| SNUNet-CD [24] | 88.16 | 78.83 | 89.18 | 87.17 | 78.57 | 64.71 | 80.06 | 77.14 | 87.27 | 77.41 | 85.45 | 89.16 |
| USSFC-Net [6] | 91.04 | 82.65* | 89.70 | **92.42** | 78.11 | 64.09 | 73.60 | 83.20 | 87.73 | 78.14 | 88.74 | 86.74 |
| HANet [26] | 90.28 | 82.27 | 91.21 | 89.36 | 76.90 | 62.47 | 78.86 | 75.04 | 88.27 | 79.01 | 87.87 | 88.68 |
| EATDer [20] | 91.20 | 81.58* | 90.67 | 91.74 | 77.15 | 62.80 | 69.10 | **87.32** | 88.06 | 78.67 | 83.75 | **92.84** |
| FEMCD [32] | 92.02 | - | **93.18** | 90.88 | 77.17 | 62.83 | 83.30 | 71.90 | 88.21 | 78.92 | 88.70 | 87.80 |
| RS-Mamba [33] | 90.10 | 82.98 | 88.23 | 92.05 | - | - | - | - | 85.24 | 75.81 | 85.43 | 85.06 |
| CDMamba [34] | 90.85 | 83.21 | 90.09 | 91.62 | - | - | - | - | 85.10 | 74.85 | 86.58 | 83.67 |
| LCCDMamba [35] | 91.68 | 84.64 | 90.96 | 92.42 | - | - | - | - | 87.14 | 77.22 | 88.37 | 85.95 |
| MVU-Net | **92.05** | **85.26** | 92.94 | 91.17 | **81.19** | **68.34** | **83.86** | 78.69 | **89.07** | **80.29** | 88.87 | 89.27 |

In the LEVIR-CD dataset, the band * indicates that the original literature has not been published, and we reproduce the results based on its public code. The public results of Changer-EX adopt the resolution of 512 × 512, while our overall comparative test adopts the resolution of 256 × 256.

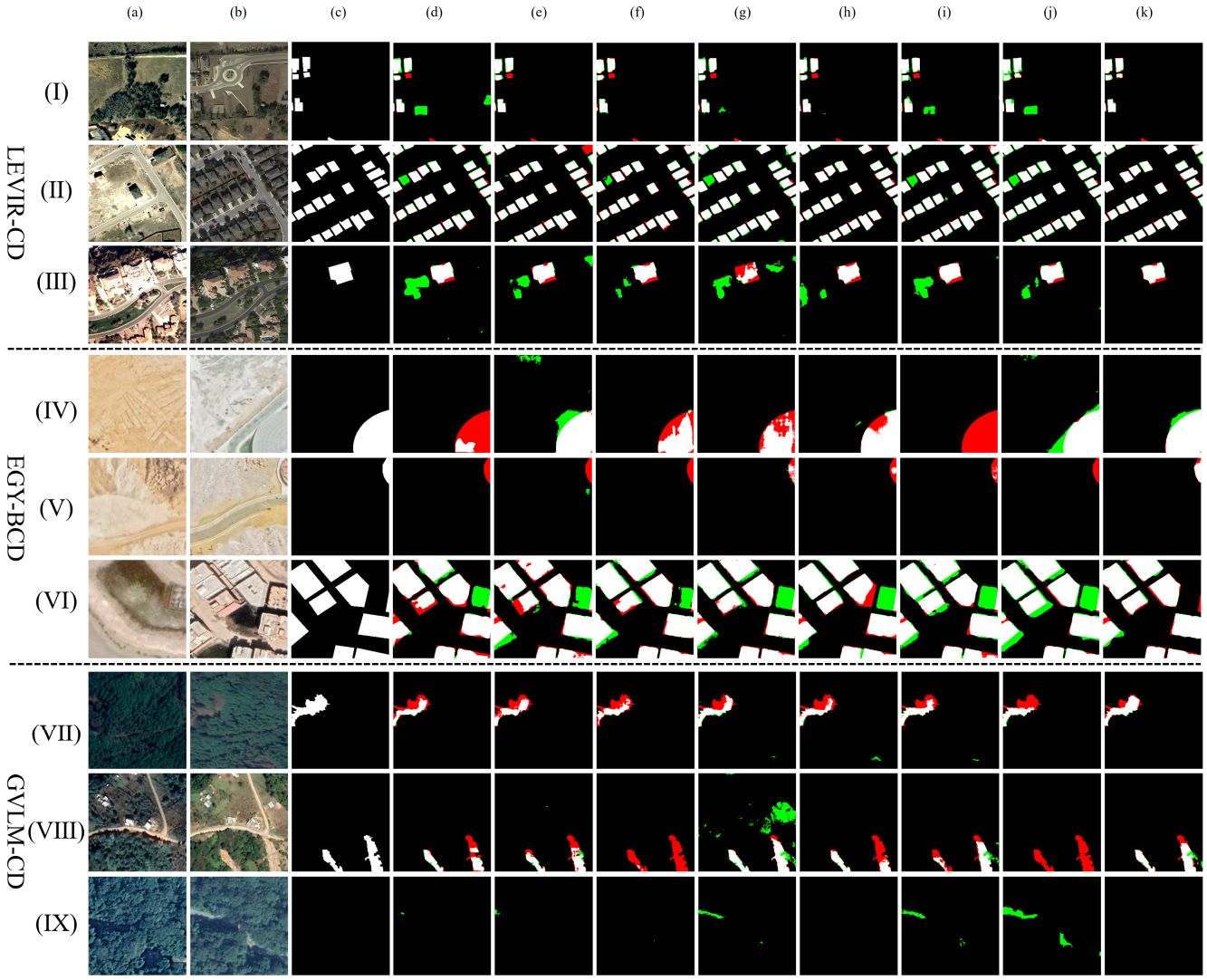


Fig. 10. Visualization results on LEVIR-CD [see (I)–(III)], EGY-BCD [see (IV)–(VI)], and GVLM-CD datasets [see (VII)–(IX)]. (a) T1 image. (b) T2 image. (c) Label. (d) BiT-CD. (e) ChangeFormer. (f) Changer-EX. (g) SNUNet-CD. (h) USSFC-Net. (i) HANet. (j) EATDer. (k) MVINet.

pairs were selected from each dataset. The resulting change diagrams, obtained by employing diverse methodologies, are presented in Fig. 10. Fig. 10(a)–(k) represents bi-temporal images, changing label images and comparison models, and

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each row Fig. 10(I)–(IX) represents different groups of remote sensing images. For ease of comparison, white is used to represent the changed target area, black is used to represent the unchanged background, red pixels are used to represent the missed detection situation, and green pixels are used to represent the false detection situation.

The visualization results of MVU-Net on the LEVIR-CD dataset are presented in Fig. 10(I)–(III). In the visualization results of Group (I), it is observed that for targets with the same foreign body spectrum, the BiT-CD, ChangeFormer, SNUNet-CD, USSFC-Net, HANet, and EATDer methods all exhibited false detection. In contrast, the results of Changer-EX and MVU-Net are similar, which can accurately avoid the impact of the target. The second building in the second row in the Group (II) image caused the false detection of BiT-CD, Changer-EX, SNUNet-CD, HANet, and EATDer models. By observing the original images, it is observed that although the building is present in both bi-temporal images, it is difficult to distinguish the building in T1. This is because strong light intensity, roof color similarity to the surrounding environment, and obvious building shadows collectively obscure its visibility. In T2, the building is different from its surroundings, leading to incorrect predictions by most methods. The images in Group (III) are also susceptible to light interference. In the original T1, the intense light causes the roof and the road surface to exhibit a strikingly similar feature representation, which impairs the model's ability to make accurate predictions. Nevertheless, even in the presence of strong light, MVU-Net can achieve superior prediction results due to our utilization of a multiperspective strategy and the diversity of change features, which effectively circumvent the influence of similar features and reduce the probability of false detections. These promising observations provide compelling evidence that MVU-Net is a practical approach.

The visualization results of MVU-Net on the EGY-BCD dataset are presented in Fig. 10(IV)–(VI). In Group (IV), attributable to the similar information representation of land and building roofs due to seasons and so on, the feature extraction ability of models, including BiT-CD and HANet, is insufficient, and it is difficult to distinguish the change regions. Meanwhile, ChangeFormer and EATDer are seriously affected by it. In contrast, the proposed MVU-Net can alleviate the above phenomenon to some extent, thanks to the fact that the MFPFM can better perceive multiscale change targets, and it can improve the integrity of building structure prediction and the clarity of building target boundaries. In the (V) Group of images, it can be seen from the results that there are many missed targets (red pixel values) in the models such as BiT-CD, ChangeFormer, and Changer-EX. In contrast, MVU-Net has better robustness against pseudo changes. This is because our proposed MVPM can effectively use multiview feature information to learn from different image perspectives, effectively use the space–time relationship between change information and features, and avoid missing detection. In addition to incomplete prediction targets and missed detections, MVU-Net can also effectively avoid the occurrence of false detections. As illustrated in the comparison images of Group (VI), intense lighting generates a large number of irrelevant shadows, leading to variations in the spectral information of unchanged regions in remote sensing images. This causes unchanged regions to be misclassified by BiT-CD, Changer-EX, ChangeFormer, SNUNet-CD, USSFC-Net, HANet, and EATDer, resulting in severe false positives. The MVPM module in MVU-Net not only effectively utilizes the multiview difference features to capture the change information and space–time relationships but also employs parallel convolution to capture the feature representations of different receptive fields. Through this feature representation, each pixel value in the feature map can obtain broader contextual information, which enables it to determine target classification by further analyzing architectural shadow features. This avoids false detections caused by shadows.

The visualization results of MVU-Net on the GVLM-CD dataset are presented in Fig. 10(VII)–(IX). In contrast to the LEVIR-CD and EGY-BCD datasets, the GVLM-CD dataset is employed to identify changes in landslide activity. In this task, the feature representation of vegetation exhibits significant variations with the change of season. The immature vegetation is easily misclassified as debris flows due to its high degree of similarity with the feature representations, which leads to the occurrence of missed detections and an increased false detection probability in CD. Furthermore, the change area in the GVLM-CD dataset is attributable to debris flow impact. Compared to the CD of buildings, the change area of this task lacks robust boundary information, which contributes to the incompleteness of the detection area. In the method comparison images of Group (VII), the shadow of the high features will also cover some vegetation and roads due to the influence of light intensity. Specifically, in the images of Group (VII), the change information affected by the light and shadow is difficult to extract from the bi-temporal image, resulting in poor detection results across all visualization methods. Group (IX) of Fig. 10 is similarly affected by light and shadow. From the bi-temporal remote sensing image pairs T1 and T2, we can observe that some roads are shaded by the shadow of trees, which leads to serious misleading and false detection of the SNUNet-CD, HANet, and EATDer methods. Overall, MVU-Net can better reduce the occurrence of false detections. This is attributed to its multiview feature extraction strategy, which efficiently models change information and spatiotemporal relationships, suppresses feature extraction from irrelevant regions, and mitigates missed detections and false positives caused by shadows. Furthermore, the similar features of various ground objects are also one of the important reasons for missed and false detections. In Group (VIII), two targets related to debris flow and landslide can be observed, but the difference lies in the fact that the target on the left is situated in the forest, whereas the target on the right is adjacent to the road. Because of the limitations in image resolution, distinguishing mud-covered roads from debris flows and landslides within the image is nearly impossible. Therefore, the Changer-EX, HANet, and EATDer methods find it difficult to determine the change features of the two types of targets, resulting in serious misses. Although the BiT-CD, ChangeFormer, and SNUNet-CD models can better predict the left side of the debris flow and landslide target, it is difficult for them to distinguish

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TABLE III
ABLATION EXPERIMENTS ON THE NUMBER AND STRUCTURE OF DISCREPANCY FEATURES ON THE LEVIR-CD, EGY-BCD, AND GVLM-CD DATASETS

| N | Type | P(M) | LEVIR-CD | | EGY-BCD | | GVLM-CD | |
|---|---|---|---|---|---|---|---|---|
| | | | F1(%) | IoU(%) | F1(%) | IoU(%) | F1(%) | IoU(%) |
| 1 | $Add$ | 3.92 | 85.96 | 75.38 | 77.4 | 63.14 | 83.49 | 71.66 |
| | $Diff$ | 3.92 | 87.14 | 77.22 | 78.38 | 64.44 | 86.17 | 75.7 |
| | $Concat$ | 4.12 | 90.8 | 83.85 | 79.79 | 66.77 | 87.38 | 78.19 |
| 2 | $Add\&Diff$ | 3.92 | 91.17 | 83.78 | 79.67 | 66.2 | 87.67 | 78.04 |
| | $Concat\&Diff$ | 4.12 | 91.69 | 84.66 | 80.49 | 67.35 | 88.18 | 78.86 |
| | $Add\&Concat$ | 4.12 | 92.05 | 85.26 | 81.19 | 68.34 | 89.07 | 80.29 |
| 3 | $Add\&Concat\&Diff$ | 9.56 | 91.83 | 84.9 | 81.77 | 69.17 | 89.28 | 80.64 |

N represents the number of different features and P represents the number of model parameters.

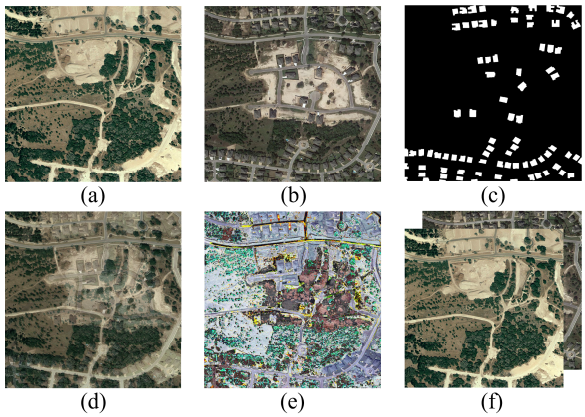


Fig. 11. Visualization of different features. (a) and (b) Bi-temporal remote sensing images, (c) real label of the change area, (d) use of add to construct different features, (e) use of diff to construct different features, and (f) use of concat to construct different features.

the right side of the mixed target effectively. It is worth noting that MVU-Net can better avoid the above problems, learn the general representation of datasets, strengthen the feature representation of changes of interest, and reduce the interference of unimportant changes.

### D. Ablation Experiment

To validate the effectiveness of the number and structure of the different features, MVPM and MFPFM, employed in our method, we performed ablation experiments on LEVIR-CD, EGY-BCD, and GVLM-CD, and reported the $F1$ and IoU results.

*1) Structure of Different Features:* As shown in Fig. 3, the different features are generated before the encoding phase of MVU-Net. In this work, we utilize the methods of "Concat" and "Add" to generate the different features. Next, we analyze the impact of the different features on the MVU-Net performance in detail. Fig. 11 shows the visualization of a pair of bi-temporal images randomly selected from the LEVIR-CD dataset and three different features. Fig. 11(d) shows the visualization results of the "Add"-based method. This method is a parameter-free processing method that can effectively highlight the change area while retaining a large amount of additional information to perceive the integrity of the change target. This is a parameter-free processing approach that effectively highlights change regions while retaining substantial auxiliary information—enabling the model to perceive the integrity of change targets. Fig. 11(e) uses the "Diff"-based method without parameters, which also highlights the change area and retains less additional information. As a common method of feature fusion, Fig. 11(f) usually "Concat"-based bi-temporal features along the channel dimension, which retains more feature information. Therefore, in the ablation experiment part, we first extracted the different features from the single-view and double-view forms of MVU-Net for comparative analysis and then verified the effect of different views on the network. The experimental results in the LEVIR-CD, EGY-BCD, and GVLM-CD datasets are presented in Table III.

By analyzing the experimental results, we come to the following three conclusions.

1) The double-perspective structure is better than the single-perspective structure. Taking the statistical results on the LEVIR-CD dataset as an illustration, under the single-view structure, $F1$ of the different features constructed in the form of add was 85.96%, but with the increase of our additional different features, $F1$ also increased significantly, and $F1$ of Add&Diff and Add&Concat increased by 5.21% and 6.1%, respectively.
2) The best effect is constructing the multiview different features using the Add&Concat method. This is because remote sensing images exhibit inherent misalignment due to varying imaging times—meaning that individual targets in the images cannot be perfectly aligned. Additionally, differences in light intensity result in variations in image intensity. Therefore, using "Diff" to construct different features inevitably introduces a lot of noise. The better solution would be to increase image registration in the preprocessing stage, which was not considered in this experiment.
3) The Add&Concat&Diff method performs better in the EGY-BCD and GVLM-CD datasets. However, the accumulation of dimensions during the feature fusion process leads to an increase in the number of model parameters, significantly reducing the training and inference efficiency of the model.

*2) Experimental Details of MVIConv Operator:* To verify the design details of the MVIConv operator, we constructed two additional interaction modes, each with different features. These are illustrated in Fig. 12. Fig. 12(a) integrates the different dimensions into the channel dimension to learn a

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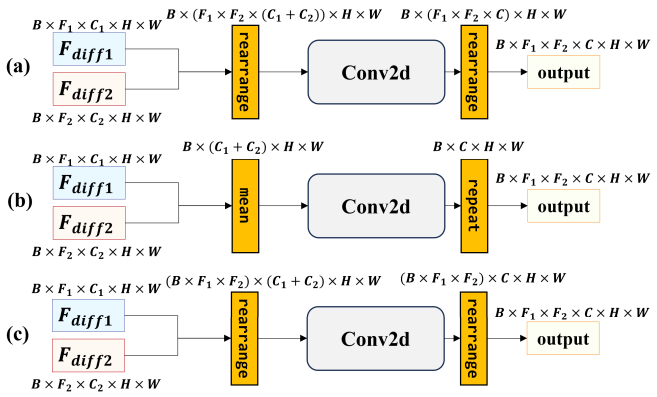


Fig. 12. Different implementations of MVIConv. (a) Integrate the different dimensions into the channel dimension to learn more comprehensive feature representation. (b) Different dimension is averaged in order to facilitate the learning of a more representative feature representation. (c) This article proposes a method for integrating the different dimensions into the batch dimension with the objective of learning more general feature representations.

TABLE IV
DIFFERENT IMPLEMENTATION DETAILS OF MVICONV OPERATOR ABLATION EXPERIMENTS ON LEVIR-CD, EGY-BCD, AND GVLM-CD DATASETS

| Type | P(M) | LEVIR-CD | | EGY-BCD | | GVLM-CD | |
|---|---|---|---|---|---|---|---|
| | | F1(%) | IoU(%) | F1(%) | IoU(%) | F1(%) | IoU(%) |
| (a) | 9.25 | 91.38 | 84.13 | 80.04 | 66.72 | 88.35 | 79.13 |
| (b) | 4.27 | 86.05 | 75.51 | 77.31 | 63.01 | 81.9 | 69.34 |
| (c) | 4.12 | 92.05 | 85.26 | 81.19 | 68.34 | 89.07 | 80.29 |

more comprehensive feature representation. Fig. 12(b) computes the mean of different dimensions, enabling the network to learn more generalized feature representations. Fig. 12(c) shows the proposed way to integrate the different dimensions into the batch dimension, which enables the learning of a more general feature representation. The experimental results are presented in Table IV. We found that the network with structure (c) exhibits fewer parameters and higher performance metrics. Compared with structure (a), the $F1$ score in the LEVIR-CD, EGY-BCD, and GVLM-CD datasets increased by 0.68% (LEVIR-CD), 1.15% (EGY-BCD), and 0.72% (GVLM-CD), respectively. Compared to structure (b), the $F1$ values increased by 6.01% (LEVIR-CD), 3.88% (EGY-BCD), and 7.17% (GVLM-CD).

*3) Effectiveness and Codec Ability of MVPM:* In order to verify the effectiveness of MVPM, we developed Base-MU-Net on the design architecture of MVU-Net, as illustrated in Fig. 13. The network structure of Base-MU-Net is analogous to that of MVU-Net. The distinction lies in the network block. Each block of MVU-Net is implemented using the MVPM module, whereas Base-MU-Net retains three stages of MVPM. The difference is that the "Concat" and split operations are used for implementation in the interaction and extraction phases. Fig. 13(a) employs splicing and splitting to facilitate the interaction and extraction of multiview features. Fig. 13(b) shows the MVPM module proposed in this article. Furthermore, to further validate the efficacy of the encoders and decoders based on MVPM, we introduced it in the encoder or decoder of the mainstream networks BiT-CD, ChangeFormer,

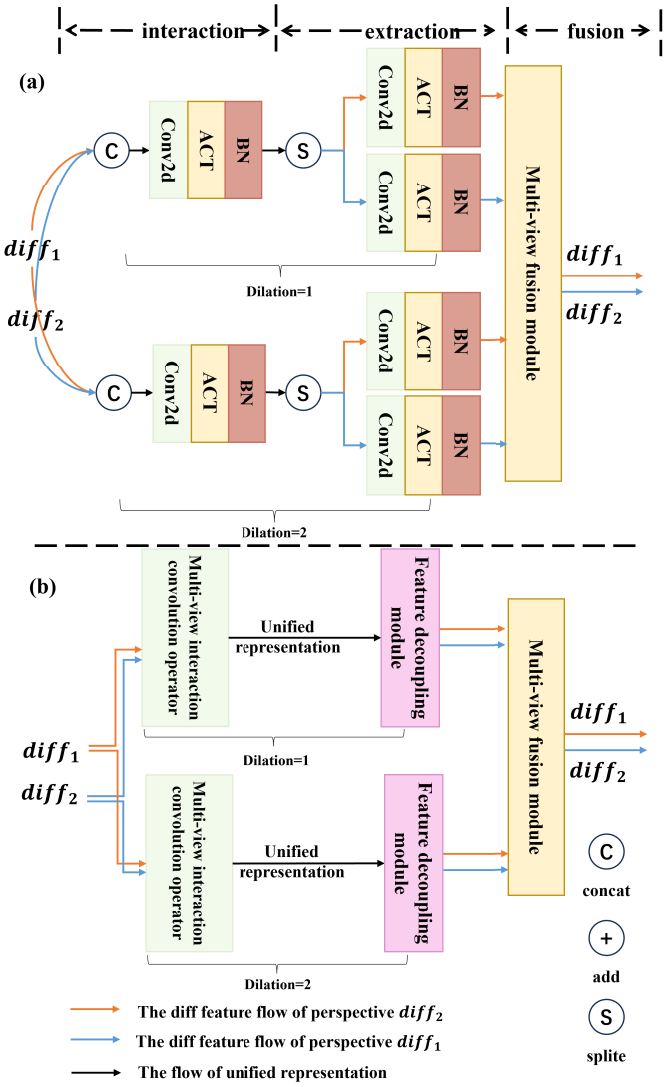


Fig. 13. (a) Design structure of block in Base-MU-Net. (b) Design structure of MVPM.

TABLE V
ABLATION EXPERIMENTS OF MVIM ENCODING AND DECODING ABILITIES ON LEVIR-CD, EGY-BCD, AND GVLM-CD DATASETS. IOU METRICS

| Encoder | Decoder | LEVIR-CD | EGY-BCD | GVLM-CD |
|---|---|---|---|---|
| BiT-CD | | 80.68% | 53.91% | 77.87% |
| BiT | MVIM | 81.94% | 63.67% | 77.51% |
| MVIM | BiT | 84.29% | 66.91% | 78.96% |
| ChangerFormer | | 82.48% | 62.92% | 77.76% |
| Changer. | MVIM | 84.67% | 67.28% | 78.88% |
| MVIM | Changer. | 83.59% | 68.08% | 80.05% |
| USSFC-Net | | 82.65% | 64.09% | 78.14% |
| USSFC. | MVIM | 82.99% | 65.6% | 76.46% |
| MVIM | USSFC. | 83.63% | 65.83% | 78.35% |
| Base-MU-Net | | 84.89% | 66.7% | 79.21% |
| MVI-Net | | 85.26% | 68.34% | 80.29% |

and USSFC-Net. Subsequently, we conducted experiments on LEVIR-CD, EGY-BCD, and GVLM-CD datasets.

The experimental outcomes are presented in Table V. Compared with Base-MU-Net, the IoU of MVU-Net in the three datasets increased by 0.4%, 1.64%, and 1.08%. The encoding

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TABLE VI

ABLATION EXPERIMENTS OF MFPFM ON LEVIR-CD, EGY-BCD, AND GVLM-CD DATASETS

| MFPFM | LEVIR-CD | | EGY-BCD | | GVLM-CD | |
|---|---|---|---|---|---|---|
| | F1(%) | IoU(%) | F1(%) | IoU(%) | F1(%) | IoU(%) |
| × | 91.68 | 84.62 | 80.53 | 67.41 | 88.5 | 79.37 |
| ✓ | 92.05 | 85.26 | 81.19 | 68.34 | 89.07 | 80.29 |

TABLE VII

IMPACT OF NETWORK DEPTH LAYER ON MODEL PERFORMANCE

| Depth | F1(%) | IoU(%) | Pre.(%) | Rec.(%) | Param. | Memory |
|---|---|---|---|---|---|---|
| 3 | 91.64 | 84.57 | 93.07 | 90.26 | 3.02 M | 14840 MB |
| 4 | 92.05 | 85.26 | 92.94 | 91.17 | 4.12 M | 15302 MB |
| 5 | 91.8 | 84.85 | 92.84 | 90.79 | 5.34 M | 15435 MB |

component of BiT-CD employs a CNN-based ResNet-18, whereas the decoding component consists of Transformer blocks. In contrast, ChangeFormer utilizes multilayer Transformer blocks in the encoding component, while the decoding component is implemented using a lightweight multilayer perceptron. USSFC-Net is a fully CNN-based network. The statistical results on the LEVIR-CD dataset demonstrate that when MVPM is applied to the encoding phase, there is an overall improvement, with an increase in IoU of 1.26% (BiT-CD), 2.19% (ChangeFormer), and 0.35% (USSFC-Net) compared to the original model. This indicates that the proposed MVPM block has a superior ability to extract change features. However, in contrast to the LEVIR-CD dataset, the GVLM-CD dataset exhibits an opposite effect. This is because, in comparison to the building CD dataset, the randomness of the change target in the GVLM-CD dataset is more pronounced, which ultimately constrains the efficacy of MVPM in the encoding phase. Conversely, when MVPM is deployed in the decoding phase, its performance is significantly superior to that when MVPM is applied to the encoding phase. On the LEVIR-CD dataset, the performance of ChangeFormer with MVPM integrated into its decoding phase is slightly inferior to that with MVPM integrated into its encoding phase. These results collectively demonstrate the effectiveness and adaptability of the proposed MVPM module.

In summary, Table V demonstrates that when MVPM is incorporated, MVU-Net achieves higher accuracy than Base-MU-Net. Additionally, integrating MVPM into the encoder or decoder enhances the CD capacity of BiT-CD, ChangerFormer, and USSFC-Net. The experimental results demonstrate that MVPM, which spans the entire network stage, improves the MVU-Net's capacity to capture the change information between the bi-temporal features. Moreover, multiview difference features are more effectively captured during both feature encoding and decoding.

*4) MFPFM Ablation Experiment:* To explore the importance of the MFPFM module (✓) in MVU-Net, we removed MFPFM (×). Subsequently, we conducted experiments on the LEVIR-CD, EGY-BCD, and GVLM-CD datasets. The $F1$ and IoU results are presented in Table VI. By comparing the results, we find that there is a significant performance difference between MVU-Net with no MFPFM and MVU-Net. Notably, compared to the original MVU-Net, the removal of MFPFM led to a decrease in $F1$ by 0.38% (LEVIR-CD), 0.66% (EGY-BCD), and 0.57% (GVLM-CD), and the decrease of IoU by 0.67% (LEVIR-CD), 0.93% (EGY-BCD), and 0.92% (GVLM-CD).

*5) Depth Layer of the MVU-Net Structure:* In this section, we conducted ablation experiments to investigate the impact of network depth, with the results presented in Table VII. The best performance is achieved when the number of layers is 4. The primary reasons for this situation are as follows.

The first reason relates to the adequacy and balance of feature extraction. With three network layers, insufficient depth hinders the comprehensive extraction of deep semantic features from remote sensing images and limits the ability to represent complex land cover features—ultimately leading to suboptimal performance in metrics such as $F1$ and IoU. With four network layers, the encoder stage effectively extracts multiscale features—retaining detailed shallow feature information (boundaries and textures) while capturing deep semantic information (feature categories and scene relationships). The decoding stage can also integrate these features more accurately, achieving a balance between detail and semantics and achieving optimal performance on measures such as $F1$ (92.05%) and IoU (85.26%).

The second reason involves the tradeoff between model complexity and generalization. When the number of layers increases to 5, the number of parameters increases significantly (from 4.12 to 5.34 M) and the memory consumption also increases to 15 435 MB. Excessive complexity not only increases the computational cost but can also lead to overfitting, increasing the model's dependence on the training data, reducing its generalization, and ultimately degrading performance (IoU drops to 84.85%). Conversely, when the number of layers is 4, the parameter count and computational complexity remain within a reasonable range—a balance that not only ensures sufficient feature extraction capability but also preserves strong generalization, thus avoiding performance loss caused by excessive complexity.

In summary, the optimal balance between feature extraction ability, computational efficiency, and model generalization is achieved when the network depth reaches a four-layer structure.

*6) Width of the MVU-Net Structure:* In this section, we conducted ablation experiments to investigate the impact of model width. The experimental results presented in Table VIII. When the network width is set to 64, the key performance indicators of the model perform well. The $F1$ reaches 92.05%, the IoU is 85.26%, and the precision and recall are 92.94% and 91.17%, respectively. Compared to the 32-channel configuration, the performance improvement is significant and the performance gap with the 96 channels is negligible. In terms of resource consumption, the 64-channel configuration has 4.12 M parameters, only 44.5% of the 96-channel configuration, effectively reducing the complexity of model storage and training; the computational efficiency reaches 15302, while ensuring the inference speed, flops is only 0.162 T, much lower than the 0.364T of the 96-channel configuration, significantly reducing the computational overhead. In summary, when the

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TABLE VIII
IMPACT OF NETWORK WIDTH ON MODEL PERFORMANCE

| Width | F1(%) | IoU(%) | Pre.(%) | Rec.(%) | Param. | Memory | Flops |
|---|---|---|---|---|---|---|---|
| 32 | 91.36 | 84.09 | 92.8 | 89.96 | 1.03 M | 7405 MB | 40.991 G |
| 64 | 92.05 | 85.26 | 92.94 | 91.17 | 4.12 M | 15302 MB | 0.162 T |
| 96 | 92.11 | 85.38 | 92.7 | 91.85 | 9.26 M | 22185 MB | 0.364 T |

TABLE IX
DISCUSSION ON PARAMETERS AND COMPUTATIONAL EFFICIENCY

| Model | Param. | FLOPs | LEVIR-CD(%) | EGY-BCD(%) | GVLM-CD(%) |
|---|---|---|---|---|---|
| FC-EF | **1.35 M** | 3.57 G | 69.07 | 51.4 | 72.51 |
| EF-Siam-conc | 1.55 M | 5.32 G | 74.5 | 52.78 | 72.46 |
| EF-Siam-diff | 1.35 M | 4.72 G | 74.5 | 52.14 | 72.1 |
| BiT-CD | 3.55 M | 10.63 G | 80.68 | 53.91 | 77.87 |
| ChangeFormer | 41.02 M | 155.25 G | 82.48 | 62.92 | 77.76 |
| Changer-EX | 3.45 M | **1.741 G** | 83.44 | 66.59 | 78.13 |
| SNUNet-CD | 12.03 M | 27.44 G | 78.33 | 64.71 | 77.41 |
| USSSFC-Net | 1.52 M | 4.86 G | 82.65 | 64.09 | 78.01 |
| HANet | 3.03 M | 14.07 G | 82.27 | 62.47 | 79.01 |
| EATDer | 6.59 M | 23.43 G | 81.58 | 62.80 | 78.67 |
| Our | 4.12 M | 162 G | **85.26** | **68.34** | **80.29** |

network width is set to 64, the model achieves the optimal balance between CD accuracy, parameter control, computational efficiency, and computational complexity.

### *E. Parameters and Computational Efficiency*

In Table IX, we systematically compared the performance of the proposed method with other mainstream CD models in terms of parameter size (Parameters), floating-point operation per second (FLOPs), and LEVIR-CD, EGY-BCD, and GVLM-CD datasets. Specifically, with a parameter size of 4.12 M, the proposed method achieved an accuracy of 85.26% on the LEVIR-CD dataset, 68.34% on the EGY-BCD dataset, and 80.29% on the GVLM-CD dataset, demonstrating better accuracy performance compared to most of the comparison models and validating the effectiveness of the method in remote sensing CD tasks. However, although its FLOPs are relatively higher than those of some lightweight models, our method has achieved substantial accuracy gains in complex scene feature extraction and change identification with a reasonable computational cost—reflecting its effective tradeoff strategy between accuracy and computational complexity.

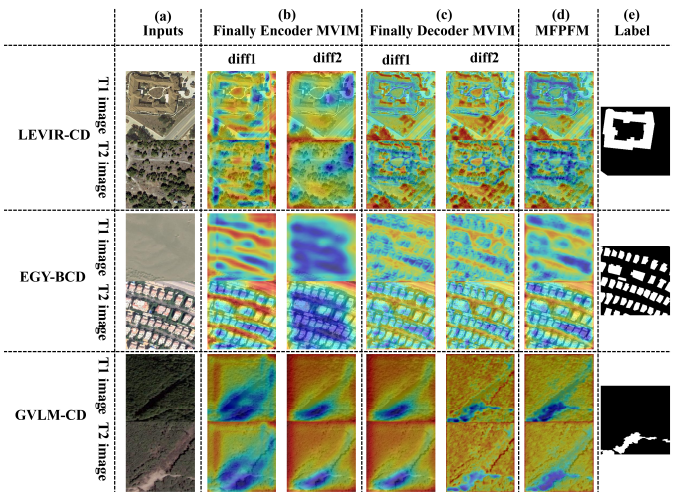


Fig. 14. Pair of dual time images selected from the LEVIR-CD, EGY-BCD, and GVLM-CD datasets is used as input network visualization examples. (a) B-temporal input image, (b) visualization of the feature map of the last MVPM of the MVU-Net encoding part, (c) visualization of the feature map of the last MVPM of the MVU-Net decoding part, (d) visualization of the feature map of the MFPFM module, and (e) real label of the bi-temporal images change.

### *F. Network Visualization*

To enable a more intuitive understanding of the functionality of different modules in MVU-Net, we present feature map visualizations at various stages of the network. Blue pixels represent the area of interest for the model, while red pixels represent the opposite, as shown in Fig. 14. This figure shows the visualization results of MVU-Net at different stages, including MVPM encoder stage, MVPM decoder stage, and MFPFM module. The visualization result is based on bi-temporal images selected from the LEVIR-CD, EGY-BCD, and GVLM-CD test sets as model inputs. For the given input image Fig. 14(a), it is first fed to the encoding part of MVU-Net. We visualize the last feature image based on the MVPM encoder. The result is shown in Fig. 14(b). Compared with the real label, we can see that the network pays more attention to the size of the modified area than to the real target. This is because in the last layer of the encoding phase, the image resolution is downsampled three times, resulting in a feature map with a wider receptive field. When visualized on the original image, the network pays more attention to a larger area. From the perspective of the different features, it is clear that different differences have different foci, but the whole is conducive to CD. The coding features are then sent to the decoder. We show the visualization of the feature map of the last MVPM decoder in Fig. 14(c). Compared with the results of the encoding part, the focus of the visualization of the decoding part is more appropriate to the real target in the image, which shows the effectiveness of the U-shaped structure. Meanwhile, the characterization from different

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perspectives proves the effectiveness of MVPM. Finally, we visualize the feature map sent to the MFPFM, as shown in Fig. 14(d). Combined with the ground truth Fig. 14(e), Fig. 14(d) exhibits fine spatial details and accurately distinguishes the changing targets. These visualization results not only enable in-depth insight into the working mechanism of MVU-Net but also confirm its robustness in handling complex CD tasks.

## VI. Conclusion

In this article, we propose a network called MVU-Net to address the challenge of change recognition ability in complex scenes for fusion features of bi-temporal images in CD tasks. Compared with current methods using a single fusion operation at a fixed stage, MVU-Net incorporates an MVPM throughout the network, enabling the interaction, extraction, and fusion of multiview difference features. Through the unified representation, decoupling, and aggregation of multiview difference features, our method is capable of capturing rich change information. Furthermore, in order to detect various scale change targets, we introduce an MFPFM. This module incrementally integrates semantic information at varying depths into the shallow texture information. Experimental results on the widely used LEVIR-CD, EGY-BCD, and GVLM-CD datasets demonstrate that the network significantly enhances the accuracy of CD, with the detection IoU reaching 85.29%, 68.34%, and 80.29%, respectively. Although this method has shown good performance on the above datasets, there are still two main limitations: 1) the computational efficiency needs to be improved, and it may face high computational requirements when processing large remote sensing data and 2) the model relies heavily on accurate registration of dual-temporal images, and in practical applications, image registration errors or differences in data acquisition conditions may affect the detection performance, leading to increased deployment difficulty. Future research will focus on lightweight model design and adaptive optimization for complex scenarios to promote their effective application in practical complex scenarios.

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